

Norm-one parameter $\rightarrow$ conductor obstruction

$q \equiv 1 \pmod{4}$, odd $n \geq 3$, $K = \mathbb{F}_q$, $Q = q^n$

$$\lambda = \alpha^q / \alpha \in T$$

$$|T| \text{ is odd} \Rightarrow \lambda = \mu^4$$

$$L_\lambda: y^2 = x(x-1)(x-\lambda)$$

Rational 2-isogeny

Kernel: O and $(0,0)$

Target root differences square

Full rational 4-torsion

16 divides one twist order

Ordinary $\Rightarrow 4$ divides f_π

Theorem applies at every size; independent of the census point counter.

Cubic extension $q = p^2$: exact absolute-Frobenius quotient

$\lambda \mapsto \lambda^p$ and $\lambda \mapsto \lambda^{-1}$ preserve the trace pair. Orbit sizes: 2, 6, 12.

One point count per orbit: $C_p = (p^4 + 3p^2 + 8)/12$ for $p > 3$.